Ganit

data speaks

Starbucks – BI & Data Engineering

Case Study

Transforming Coffee Retail Analytics with Unified BI & Scalable Data Engineering on Cloud

One Pager

Client	Domain	Stack	Outcome
Starbucks Corporation	**BI + Data Engineering**	**Azure + Power BI + dbt**	**60% Faster Decisions**
Global Coffee Retail & QSR	Cloud Analytics Platform	Databricks · ADF · Synapse	1M+ transactions/day unified

Business Intelligence

- Unified sales, loyalty & operations KPIs in a single Power BI platform
- Store-level P&L, regional performance & menu analytics for 35,000+ outlets
- Real-time loyalty redemption & campaign ROI dashboards
- Row-level security for region/country managers

Data Engineering

- Azure Data Factory pipelines ingesting POS, mobile app & supply chain data
- Medallion architecture (Bronze → Silver → Gold) on Azure Data Lake Gen2
- dbt-powered transformations with automated data quality checks
- Near real-time streaming via Azure Event Hubs + Databricks Delta Live Tables

Unifying Starbucks' Retail & Loyalty Analytics with Cloud-Native BI & Data Engineering — Enabling Real-Time Decisions Across Sales, Supply Chain & Loyalty

Problem Statement

- No unified view of POS, loyalty & supply chain
- Siloed data across 35,000+ stores globally
- Manual reporting with 48-hour lag
- Legacy on-prem DWH causing high infra costs

Solution Overview

A leading global coffee brand needed a unified analytics platform ingesting POS transactions, Starbucks Rewards loyalty data, and supply chain feeds to drive decisions across 35,000+ outlets worldwide.

Application UI

- Power BI Embedded dashboards
- Store-level sales & P&L views
- Loyalty & campaign performance

Business Value Achieved

✓ 40% faster menu & pricing decisions
✓ Eliminated manual reporting bottlenecks
✓ Migrated 5+ TB of historical POS & loyalty data
✓ Reporting latency cut from 48 hrs to under 2 hours
✓ Real-time loyalty KPIs for 30M+ members
✓ Data costs reduced by ~70% vs on-prem DWH

Solution Proposed by Ganit

POS / Mobile App / Supply Chain APIs → Azure Data Factory + Event Hubs → ADLS Gen2 (Bronze → Silver → Gold) → Azure Synapse + Databricks → Power BI Semantic Model → Power BI Dashboards

Azure Services Used: ADLS Gen2 | Azure Data Factory | Event Hubs | Databricks | Azure Synapse | Power BI Premium | dbt | Azure Purview

Ganit unified Starbucks' retail, loyalty & supply chain data on Azure, accelerating analytics adoption and driving decisions across all global markets

Outcome

Delivered a cloud-native Lakehouse and semantic layer on Azure, ingesting data from POS systems, Starbucks Rewards mobile app, and supply chain REST APIs through automated ADF Pipelines — surfaced through Power BI for self-service analytics across all global markets.

Operational Efficiency

Eliminated manual file-handling and siloed POS exports by automating ingestion via ADF Pipelines and Databricks Delta Live Tables, reducing data latency from 48 hours to under 2 hours.

Cost Optimization

Used ADLS Gen2 intelligent tiering and Azure Synapse serverless for infrequent historical queries, achieving ~70% reduction in legacy on-prem DWH costs while scaling on demand.

Scalability & Reliability

Built a multi-zone Medallion architecture (Bronze / Silver / Gold) with Purview governance, handling 5+ TB of historical data and supporting 200+ concurrent Power BI users.

Business Growth

Enabled real-time store performance, loyalty ROI, and supply chain dashboards in Power BI, giving regional teams actionable visibility into menu-level sell-through and loyalty redemption rates.

Fragmented POS, loyalty, and supply chain data blocked unified retail analytics across Starbucks' global markets

Problem Statement

Starbucks operates 35,000+ outlets globally with sales spanning in-store POS, the Starbucks Rewards mobile app, and third-party delivery integrations.
Data was siloed across Oracle POS systems, the Starbucks Rewards CRM, manually uploaded store performance Excel files, and supply chain ERP feeds — with no automated pipeline connecting these to a unified analytics layer. This led to delayed decisions, inconsistent loyalty reporting, and an inability to measure menu-level ROI in near-real time.

Challenge 1	Fragmented POS & loyalty data with no automated cross-system pipeline
Challenge 2	Manual Excel uploads causing 48-hour reporting delays across regions
Challenge 3	Supply chain ERP & delivery partner API data stranded in legacy systems
Challenge 4	No unified SKU-level view of sell-through vs loyalty redemption by market

Ganit built a cloud-native BI & Data Engineering platform on Azure with automated ingestion from POS, Loyalty & Supply Chain — served through Power BI for real-time retail analytics

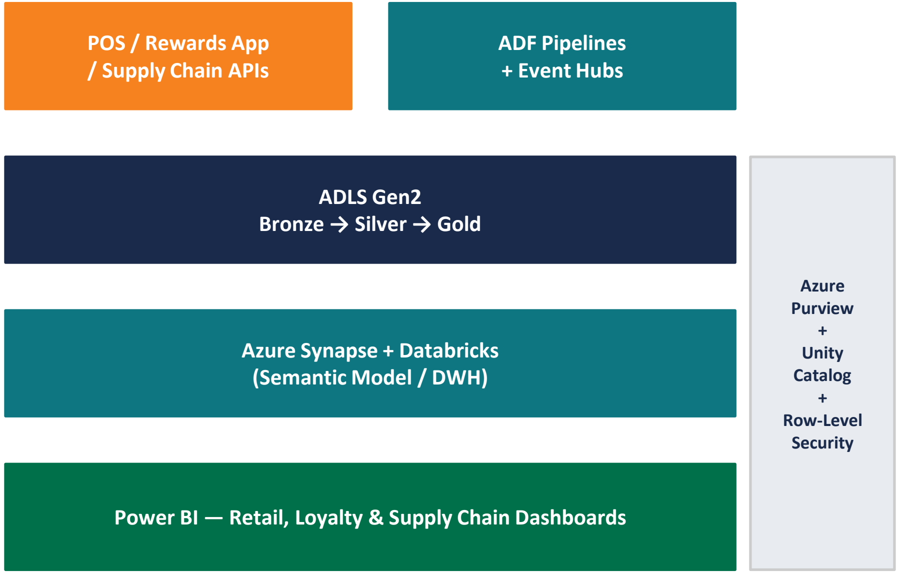
Solution Overview

Robust & Scalable Architecture

- POS data ingested via ADF Pipelines + custom connectors with incremental delta loads.
- Starbucks Rewards app events streamed via Azure Event Hubs + Databricks Delta Live Tables.
- Supply chain ERP & delivery REST APIs polled on schedule via ADF with retry logic.
- Bronze → Silver → Gold ADLS Gen2 zones with schema enforcement via Unity Catalog.

Application UI — Power BI on Azure

- Deployed Power BI Premium Embedded with live connections to Azure Synapse Warehouse.
- Dashboards for Sales Performance, Loyalty Analytics, Supply Chain & Menu Profitability.
- Row-level security enforced via Azure AD groups mapped to regional market managers.



6 Dashboards | 150+ KPIs | 5+ TB Historical Data | 200+ Concurrent Users | 5 Source Systems | 1M+ Transactions/Day